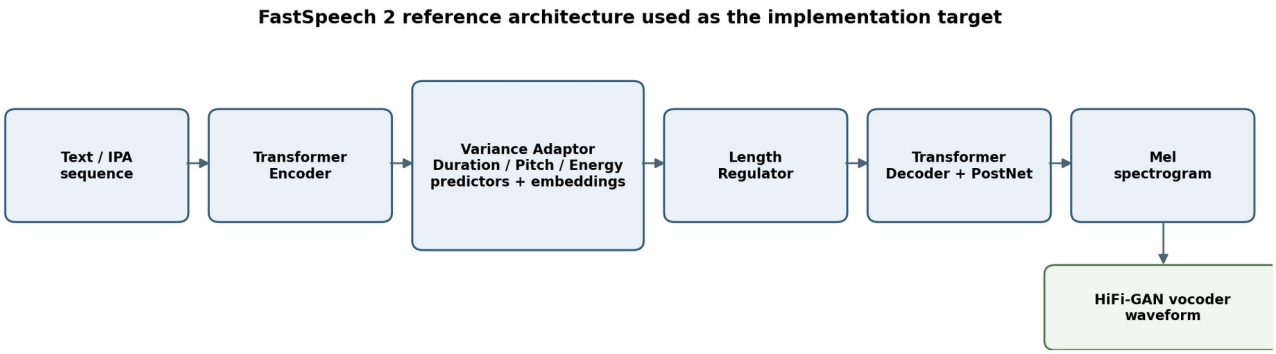
TECHNICAL REPORT

Implementation of FastSpeech 2-Based Text-to-Speech Synthesis

English (Svarah) and Telugu (IndicTTS)



INTERIM IMPLEMENTATION STATUS AS OF 07 NOVEMBER 2026

Prepared against the assignment brief dated 07-11-2026

Status and evidence note

This report records only work supported by the executed notebook outputs, logs, configuration snapshots and alignment test output available at the report date. It does not claim completed Telugu FastSpeech 2 training, full-corpus Telugu alignment, generated test audios, loss curves or variance-adaptor plots where those artifacts have not yet been produced.

Executive Summary

The assignment requires two separate FastSpeech 2 systems, one for English using Svarah and one for Telugu using IndicTTS Telugu, together with test-sentence synthesis and analysis of loss curves and variance-adaptor predictions. The implementation began from the public ming024/FastSpeech2 reference repository and retained separate language data, configuration, vocabulary and output paths.

The English pipeline is substantially further along: the selected Svarah manifests were prepared, MFA alignments and acoustic features were generated for most records, the converted arrays passed integrity checks, the reference dataset loader and model forward path were smoke-tested, one optimizer update completed with finite gradients, and a HiFi-GAN vocoder checkpoint loaded and returned finite waveform output. A complete long-running FastSpeech 2 training run and assignment test-sentence evaluation are not evidenced in the current artifacts.

The Telugu pipeline reused the existing IndicTTS local data and the existing 5-hour/1-hour split. All 1,857 audio paths and WAV headers were verified, working WAV/LAB files were exported at 22.05 kHz, and an eSpeak NG Telugu candidate lexicon was generated. MFA 3.4.3 passed structural validation. Its acoustic training progressed through SAT2 and produced a recoverable model archive, but the command failed in the final pronunciation-probability calculation. The recovered model package passed ZIP integrity and MFA inspection, and one Telugu utterance successfully produced a TextGrid using align_one.

Overall status: Data preparation and alignment infrastructure are established for both languages, but the assignment is not yet fully complete. Full Telugu corpus alignment, Telugu FastSpeech 2 feature serialization/training/inference, both languages' final test audios, final loss curves and variance-adaptor plots remain outstanding.

Implementation Status Snapshot

Requirement area	English	Telugu
Dataset and split	Svarah manifests: 2,908 train / 553 validation rows.	IndicTTS: 1,544 train / 313 validation; 5.0031 h / 1.0043 h; zero overlap.
Audio preparation	22.05-kHz working audio and alignment corpus prepared.	1,857 WAV/LAB pairs exported at 22.05 kHz; source data retained.
Alignment	2,900/2,908 train and 548/553 validation TextGrids.	MFA structural validation passed; one align_one TextGrid passed; full alignment pending.
FS2 features	2,897 train and 548 validation feature arrays passed audit.	Full mel/pitch/energy/duration arrays not yet generated.
FS2 training	Forward/backward and one optimizer update tested; full training not evidenced.	Not started; no Telugu FS2 checkpoint.
Inference / plots	Required final test audio and variance plots not evidenced.	Not generated.

1. Objective and Implementation Approach

The implementation target is the non-autoregressive FastSpeech 2 architecture. The reference code provides the Transformer encoder and decoder, variance adaptor, duration-based length regulator, PostNet, training/evaluation loops, preprocessing utilities and vocoder integration. The same broad architecture and acoustic settings were used as the starting point for the two languages, while language-specific datasets and text frontends were kept separate.

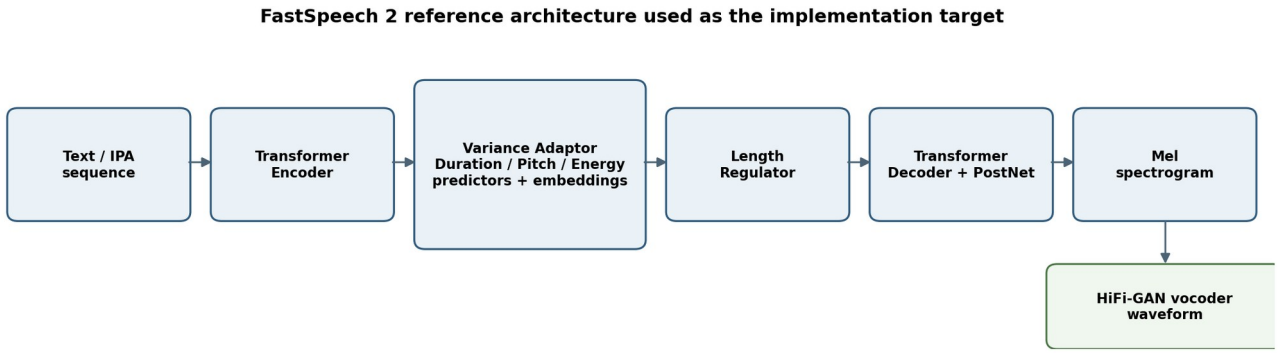


Figure 1. FastSpeech 2 data flow. The variance adaptor predicts duration, pitch and energy; the length regulator expands phoneme representations to frame-level sequence length before mel decoding. HiFi-GAN converts the predicted mel spectrogram into a waveform.

1.1 Reference implementation components

- Text frontend and embedding: convert the input text/phoneme sequence into token IDs and embeddings.
- Transformer encoder: four layers, two attention heads and 256 hidden units in the current configuration.
- Variance adaptor: duration, pitch (F0) and energy predictors; pitch/energy embeddings; and a length regulator.
- Transformer decoder and PostNet: six decoder layers, two heads and 256 hidden units, followed by mel refinement.
- HiFi-GAN vocoder: converts mel spectrograms to audio. The Telugu configuration selects the universal HiFi-GAN checkpoint family when available.

1.2 Language-specific modifications

Area	English / Svarah	Telugu / IndicTTS
Data reader	Manifest-based Svarah WAV and transcript mapping; normalized transcript columns retained for audit.	Local-first IndicTTS reader; uses existing train/validation manifests and Telugu-only raw audio lookup.
Text normalization	English transcript normalization versions were compared; numeric/date and identifier cases were reviewed.	Unicode NFC and whitespace normalization only; source orthography preserved.
Audio	Mono 22.05-kHz working audio; English MFA corpus and TextGrid tiers.	Source 16/48-kHz mono WAVs resampled to 22.05 kHz; source files left unchanged.
Phone frontend	English ARPAbet dictionary and language-specific symbols.	eSpeak NG Telugu IPA candidate dictionary, manual override template and Telugu-specific symbol extension script.
Speakers	Single speaker ID in the English alignment/training conversion.	Two speaker labels (`female`, `male`) and planned multi-speaker FastSpeech 2 configuration.

2. Dataset Selection, Subsampling and Preprocessing

2.1 English: Svarah

The English pipeline used the provided Svarah training and validation manifests and resolved their dataset indices to local WAV files. The manifest snapshots contained 2,908 training rows and 553 validation rows. MFA produced 2,900 training TextGrids and 548 validation TextGrids. The combined alignment coverage was 3,448 of 3,461 utterances (99.62%); 8 training and 5 validation utterances remained without TextGrids.

English dataset stage	Training	Validation

Manifest rows / audio records	2,908	553
TextGrids	2,900	548
Final feature NPZ files	2,897	548
Feature files passing integrity	2,897	548
Missing alignment	8	5
Missing features after alignment	3	0

The English transcript workflow retained multiple normalization versions and a final transcript field. Number normalization and other transcript edits were reviewed through preview manifests rather than applied blindly. For example, one preview records 402 changed training rows in a normalization pass; the final data manifest retained the final transcript string used for phone conversion. The English split was maintained as separate train and validation manifests; the overlap audit in the preprocessing workflow was zero.

The English FastSpeech 2 feature configuration used 22,050 Hz audio, 1,024-point FFT/window, 256-sample hop, 80 mel channels and an 8,000-Hz mel upper bound. Pitch and energy were phoneme-level features. The feature conversion used training-only normalization statistics: pitch mean 177.8983 and standard deviation 56.8179; energy mean 27.3366 and standard deviation 26.1940.

English aggregate	Training	Validation
Feature records	2,897	548
Total phonemes	157,800	31,895
Total mel frames	1,424,387	287,344
Feature integrity	All 2,897 passed	All 548 passed

2.2 Telugu: IndicTTS

The Telugu notebook did not re-download the corpus. It reused the local `data/manifests/telugu_train.csv` and `telugu_validation.csv` files and resolved their `dataset_index` values against the existing `data/raw/Telugu/{train,validation}` directory layout. The already-created train/validation partition was reused because it was present, near the target duration and identifier-disjoint.

Telugu split	Utterances	Duration (hours)	Speakers
Training	1,544	5.003101	female 596; male 948
Validation	313	1.004286	female 115; male 198
Total	1,857	6.007387	two speaker labels

The audio audit resolved all 1,857 manifest records to audio, with no missing WAVs. All 1,857 WAV headers were readable and there were no sample-rate mismatches against the audit metadata. The source sample-rate distribution was 1,146 recordings at 48 kHz and 711 at 16 kHz. The source files were mono. Working copies were resampled to 22,050 Hz using polyphase resampling and written as PCM-16 WAV files; text was normalized with Unicode NFC and whitespace normalization only. Original files under `data/raw` were not overwritten.

- MFA corpus layout: `mfa_corpus/train/<speaker>/<utterance>.wav` plus a same-basename `.lab` transcript; equivalent layout for validation.
- A file-level check confirmed 1,544 training WAV/LAB pairs and 313 validation WAV/LAB pairs. Train and validation IDs were disjoint.
- Seventeen rows were reported in the duplicate-transcript diagnostic. They were not automatically removed because distinct speakers/recordings can legitimately share the same utterance text.

3. Telugu Text Frontend, Dictionary and Forced Alignment

3.1 Candidate dictionary

The notebook tokenized the combined training and validation transcripts and used the eSpeak NG Telugu voice (`te`) through Phonemizer to produce candidate IPA phone sequences with explicit phone separators. It reported 11,788 unique training words, 3,308 unique validation words, 1,673 validation-only words and 13,461 unique words in the combined vocabulary. The candidate dictionary contains 13,461 entries, 47 distinct candidate phone symbols and zero parser-reported G2P errors. A manual override CSV was created, but zero overrides had been applied at the current report checkpoint.

Pronunciation quality limitation: The dictionary is structurally valid but not fully human-verified. A standalone `ʲ` token occurs in 234 entries (examples include words beginning with `ఎ`). The fact that MFA accepts a phone token does not prove that the tokenization or pronunciation is linguistically correct. Manual review remains required.

3.2 MFA environment and validation

Montreal Forced Aligner 3.4.3 was installed in an isolated environment at `/content/mfa-env`; OpenFST was confirmed at `/content/mfa-env/bin/fstcompile`. The structural validation command was run with `--ignore_acoustics`. It exited with code 0 and reported 1,544 sound files, 1,544 text files, two speakers, no sound-file read errors, no missing transcriptions and no out-of-vocabulary words. Acoustic feature generation and trial alignment were explicitly skipped in that first structural preflight.

The subsequent MFA acoustic-model training run completed the monophone stage, 35 triphone iterations, 35 LDA+MLLT iterations, 35 SAT iterations and 35 SAT2 iterations. It then failed during pronunciation-probability estimation with `ZeroDivisionError: division by zero` while computing silence probability. Thus, the standard `mfa train` command did not exit successfully and did not produce the intended final top-level model and training TextGrid export.

A SAT2 package was preserved from `sat_2/log/acoustic_model.zip` and copied to Drive as `telugu_acoustic_model_sat2_recovered.zip`. The ZIP integrity test passed. `mfa model inspect acoustic` returned code 0 and described a 3.4.3 GMM-HMM triphone model using MFCCs, 10-ms frame shift, LDA and speaker adaptation. Its phone list contained the 47 candidate phone symbols, including `ʲ`.

The first alignment attempt failed before processing audio because `fstcompile` was not discoverable on PATH. After confirming OpenFST in the MFA environment and adding `/content/mfa-env/bin` to PATH, the same one-utterance test succeeded. `mfa align_one` aligned `female/indictts_00002.wav` against its `.lab` transcript and created a 27,305-byte long-format TextGrid with return code 0. Full training and validation corpus alignment have not yet been run.

4. FastSpeech 2 Configuration and Training Evidence

4.1 Reference configuration

Component	Current configuration
Encoder	4 Transformer layers; 2 heads; hidden size 256; dropout 0.2
Decoder	6 Transformer layers; 2 heads; hidden size 256; dropout 0.2
Convolutional feed-forward	Filter size 1,024; kernels [9, 1]
Variance predictors	Filter size 256; kernel size 3; dropout 0.5
Variance embeddings	Linear pitch and energy quantization; 256 bins
Sequence length	Maximum sequence length 1,000
Audio and mel	22,050 Hz; 80 channels; FFT/window 1,024; hop 256; mel 0-8,000

	Hz
Optimizer pilot	Adam; batch size 16; betas (0.9, 0.98); gradient clip 1.0; warm-up 1,000 steps; 5,000-step pilot
Vocoder	HiFi-GAN; universal checkpoint family selected for Telugu configuration

The Telugu configuration is marked multi-speaker and uses a deterministic speaker map based on the labels `female` and `male`. The Telugu model must use its own phone vocabulary and embeddings; the English checkpoint and English symbol indices must not be reused for Telugu.

4.2 English model smoke tests

The English implementation was tested at several layers. A five-recording conversion smoke test generated mel, pitch, energy and duration arrays and all five records passed the custom checks. The reference Dataset loader successfully loaded five records, returned the expected eight dictionary elements, and a separate batch test transferred the tensors to the NVIDIA L4 GPU. A FastSpeech 2 forward pass and loss computation produced finite values.

Loss component	One forward-pass smoke test
Total loss	19.405998
Mel loss	5.483802
Mel PostNet loss	5.524628
Pitch loss	2.014973
Energy loss	1.467228
Duration loss	4.915366

A separate one-step optimization test completed backpropagation and one optimizer update. It reported 215 parameter tensors with gradients, all gradients finite, gradient norm 61.539 before clipping, and finite parameter values after the update. These are smoke-test results, not a convergence curve or a completed training run.

The HiFi-GAN generator checkpoint was loaded successfully on the L4. A synthetic mel input of shape (1, 80, 100) produced finite waveform output of shape (1, 1, 25,600). This verifies the vocoder forward path only; it is not a Telugu or English test-sentence synthesis result.

4.3 Telugu model training status

The recovered MFA SAT2 model is an acoustic aligner, not a FastSpeech 2 neural checkpoint. Telugu FastSpeech 2 training has not started because full training and validation TextGrids have not yet been generated, and the phone-duration/mel/pitch/energy arrays and `train.txt`/`val.txt` metadata have not yet been written. The 5,000-step, batch-16 configuration in the Telugu folder is a planned pilot configuration only.

5. Training and Validation Loss Analysis

Loss-curve availability: No complete English or Telugu FastSpeech 2 training/validation loss curves were present in the artifacts reviewed for this report. The English one-batch and one-optimizer-step values are isolated smoke tests. Telugu FastSpeech 2 loss values do not exist yet. Therefore convergence, instability and overfitting cannot be assessed from the current evidence.

Once training is run, the report should plot total loss and the five component losses (mel, mel PostNet, pitch, energy and duration) at the logged steps, with the validation curve on the same step axis. A falling training loss accompanied by flat or rising validation loss would indicate possible overfitting; persistent oscillation or

non-finite values would indicate instability. Those observations are a future analysis plan, not findings from the present runs.

6. Generated Speech and Variance-Adaptor Analysis

6.1 Required test-sentence speech

The assignment specifies five English and five Telugu evaluation items (E2 has two sub-sentences; T2 has two sub-sentences). The current evidence does not include a generated WAV file for any of the required test sentences. Consequently, pronunciation accuracy, intelligibility, naturalness, prosody, speaking rate and overall audio quality cannot yet be judged. The HiFi-GAN synthetic-mel smoke test is not a substitute for these evaluations.

6.2 Duration, pitch and energy plots

No assignment test-sentence variance-adaptor plots have been generated for either language. The English preprocessing work contains phoneme-level duration, pitch and energy targets and the reference model exposes corresponding predictors, but test-sentence predictions require a trained FastSpeech 2 checkpoint and inference output. Telugu feature targets are still blocked on full TextGrid alignment.

- Duration plots should show predicted duration per input phone/token and identify long pauses, unusually short phones and long compound words.
- Pitch plots should show the predicted F0 contour in Hz (or clearly labeled normalized scale) and identify unvoiced/zero regions, phrase-final movement and implausible jumps.
- Energy plots should show per-phone predicted energy and relate peaks to syllabic prominence, stress and phrase boundaries.
- Each plot should be tied to the exact assignment sentence ID and accompanied by listening observations from its generated WAV.

7. English–Telugu Comparative Observations

Dimension	English	Telugu
Dataset readiness	Manifest and audio mapping complete; 3,461 total manifest rows.	Manifest/audio audit complete; existing 5h/1h split reused.
Alignment	3,448 TextGrids across train/validation; 13 exclusions.	MFA trained through SAT2; recovered package; one utterance aligned.
Text frontend	English normalization and ARPAbet path.	Telugu Unicode/NFC text and eSpeak IPA candidate lexicon; 234 `ఁ` entries need review.
Feature targets	3,445 NPZ feature records passed integrity checks.	No full FS2 feature arrays yet.
FS2 training/inference	Forward/backward and one update tested; complete model training/inference not evidenced.	Not started; no neural checkpoint or generated audio.

The English workflow is closer to the model-training boundary because the full feature arrays have been serialized and passed compatibility checks. Telugu has a clean local data split and a recovered language-specific MFA acoustic checkpoint, but it is still upstream of FastSpeech 2 training. A direct speech-quality comparison is therefore not yet possible. The main Telugu-specific risk is the quality and consistency of the IPA phone inventory; the main English limitation in the current evidence is the small number of missing alignments/features and the absence of a completed neural training/evaluation run.

8. Limitations, Failures and Planned Improvements

Issue	Observed impact	Action

MFA final probability error	MFA train command exited with ZeroDivisionError after SAT2; no standard final export.	Preserve SAT2 archive; use it for smoke/full alignment tests; investigate dictionary registration separately.
OpenFST PATH	Initial align_one failed before audio processing because `fstcompile` was not found.	Add `/content/mfa-env/bin` to PATH; the subsequent single-file alignment passed.
Unreviewed G2P	Candidate dictionary has zero parser errors but is not fully human-verified; 234 entries include standalone `ఀ`.	Review common and long words, names, code switches and modifiers with a Telugu reader; record overrides.
Missing full Telugu TextGrids	No phone-duration targets for all 1,544/313 utterances yet.	Run align-train and align-validation; audit tier names, interval continuity and coverage.
No Telugu FS2 features/training	No Telugu neural checkpoint, losses, generated samples or variance plots.	Run reference TacotronSTFT-based feature extraction; validate duration/mel equality; then train and evaluate.
English missing alignments/features	8 train + 5 validation alignments missing; 3 train feature files missing after alignment.	Document exclusions or retry, then verify the final metadata counts and feature coverage.

9. Reproducibility and Commands

9.1 Recorded environment

Component	Recorded value
Runtime	Google Colab Linux x86_64
Python in Colab	3.13.16
PyTorch	2.14.1+cu130
GPU	NVIDIA L4, 23,034 MiB reported
MFA	3.4.3 in isolated `/content/mfa-env`
OpenFST	`/content/mfa-env/bin/fstcompile`
Audio target	22,050 Hz, mono, PCM-16
Acoustic settings	80 mel channels; FFT/window 1,024; hop 256; 0-8,000 Hz

9.2 Telugu command sequence

```
# Run from the project root; source manifests/audio must already exist.
python Telugu/preprocess.py \
  --train-manifest data/manifests/telugu_train.csv \
  --validation-manifest data/manifests/telugu_validation.csv \
  --raw-root data/raw/Telugu \
  --output-root data/processed/telugu

# Candidate G2P dictionary and review files
python Telugu/build_lexicon.py

# Set the MFA executable and aligner environment
export MFA_BIN=/content/mfa-env/bin/mfa
python Telugu/align.py validate
python Telugu/align.py smoke-test
python Telugu/align.py align-train --num-jobs 2
python Telugu/align.py align-validation --num-jobs 2

# After all TextGrids exist
```

```
python Telugu/extract_features.py --repo-root .
python Telugu/patch_fastspeech2_symbols.py --repo-root .

# Preflight first, then explicitly launch the reference trainer
python Telugu/train.py --repo-root .
python Telugu/train.py --repo-root . --execute

# After a checkpoint exists
python Telugu/inference.py --repo-root . \
  --text "న్యూరోఫార్మాకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు." \
  --speaker female --restore-step 5000
```

9.3 Configuration files

- `Telugu/config.yaml`: data paths, split snapshot, audio parameters, speaker labels, MFA executable/model paths and planned FastSpeech 2 pilot settings.
- `Telugu/configs/preprocess.yaml`: reference FastSpeech 2 preprocessing schema and data/feature settings.
- `Telugu/configs/model.yaml`: Transformer, variance predictor, embedding and vocoder configuration.
- `Telugu/configs/train.yaml`: optimizer, batch size, warm-up, logging/evaluation and checkpoint intervals.
- `Telugu/test_sentences.json`: exact assignment Telugu evaluation inputs; these are not training records.

9.4 Repository and licensing practice

Keep the original reference implementation and its license/attribution. Keep English and Telugu code/config/checkpoints separate. Do not upload raw IndicTTS speech or processed WAV/TextGrid/feature folders to a public GitHub repository unless the dataset license explicitly permits redistribution. Put large checkpoints in Git LFS or provide an access-controlled download link in README. A recovered MFA acoustic model and a Telugu FastSpeech 2 neural checkpoint are different artifacts and should have separate names.

10. Conclusion and Remaining Work

The English workflow has completed dataset mapping, alignment for the large majority of records, acoustic feature serialization, dataset compatibility checks and initial model/vocoder smoke tests. The Telugu workflow has completed local data verification, the 5-hour/1-hour split, 22.05-kHz WAV/LAB export, candidate dictionary generation, MFA structural validation, recovery of an MFA SAT2 package and one successful forced-alignment test. These are meaningful implementation milestones.

The deliverables that require trained FastSpeech 2 models are not yet complete. The critical Telugu next step is full train and validation alignment with the recovered MFA model, followed by audited phone-duration extraction and reference-compatible mel/pitch/energy serialization. Only after those data pass integrity checks should Telugu FastSpeech 2 training and the assignment's ten sentence-level inference/plot analyses be run. English likewise needs its complete neural training and final sentence-level synthesis/evaluation evidence.

Submission statement: This is an interim technical report. It documents the work actually completed and identifies pending deliverables; it should be updated with final loss curves, checkpoints, generated WAVs, listening notes and variance-adaptor plots before it is presented as a final experimental report.

Appendix A. Assignment Evaluation Sentences

The strings below are transcribed from the assignment brief and retained as evaluation inputs. T2 and E2 each contain two sub-sentences, giving five test items per language.

A.1 English

ID	Exact test text
E1	The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.
E2a	The orchestra's conduct during the ceremony was praised by the audience.
E2b	The researchers will conduct an experiment to evaluate the proposed method.
E3	The rhyth m of the orchestra changed unexpectedly.
E4	The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.

A.2 Telugu

ID	Exact test text
T1	న్యూరోఫార్మాకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.
T2a	తాడుపై అతని పట్టు చాలా బలంగా ఉంది.
T2b	ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.
T3	పులి అడవిలో నిశ్శబ్దంగా నడిచింది.
T4	ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.

Appendix B. Artifact Checklist at Report Date

Artifact	Status at 07 Nov 2026
English preprocessing and feature arrays	Available; 3,445 NPZ files pass integrity checks.
English training/validation loss plots	Not evidenced.
English trained FastSpeech 2 checkpoint	Not evidenced as a completed long training run.
English five test WAV files	Not evidenced.
English duration/pitch/energy plots	Not evidenced.
Telugu source/processed manifests and WAV/LAB corpus	Available on the user Drive; 1,857 records.
Telugu candidate lexicon / override sheet	Available on Drive; manual review incomplete.
Telugu MFA recovered SAT2 archive	Available on Drive; inspected; not a final successful MFA train export.
Telugu one-utterance TextGrid	Available on Drive; align_one smoke test passed.
Telugu full train/validation TextGrids	Pending.
Telugu FastSpeech 2 arrays, checkpoint, loss plots	Pending.
Telugu five test WAV files and variance plots	Pending.

References

1. Ren, Y., Hu, C., Tan, X., Qin, T., Zhao, S., Zhao, Z., and Liu, T.-Y. (2021; arXiv:2006.04558). FastSpeech 2: Fast and High-Quality End-to-End Text to Speech. https://arxiv.org/abs/2006.04558
2. ming024. FastSpeech2: PyTorch implementation of FastSpeech 2. https://github.com/ming024/FastSpeech2
3. Montreal Forced Aligner documentation, version 3.x. https://montreal-forced-aligner.readthedocs.io/
4. SPRINGLab. IndicTTS Telugu dataset. https://huggingface.co/datasets/SPRINGLab/IndicTTS_Telugu
5. Assignment brief: Implementation of FastSpeech 2-Based Text-to-Speech Synthesis, deadline 07-11-2026 (user-provided PDF).